

		MIS 2	MIS 6	MIS 7d	MIS 8	MIS 10	MIS 12	MIS 14	MIS 15b	MIS 16	MIS 18	MIS 20	MIS 22	MIS 26	MIS 28	MIS 30	MIS 32	MIS 34	MIS 36	MIS 38	MIS 40	MIS 42	MIS 44	MIS 46	MIS 48
Benthic $\delta^{18}\text{O}$ (‰)	Lisiecki & Raymo (2005)	5.02	4.98	4.44	4.63	4.84	5.08	4.55	4.33	5.08	4.75	4.74	4.69	4.58	4.28	4.54	4.29	4.50	4.43	4.36	4.35	4.19	4.26	4.40	4.31
	Ahn et al. (2017)	4.94	4.91	4.43	4.63	4.83	5.09	4.68	4.35	5.14	4.64	4.72	4.72	4.51	4.35	4.43	4.30	4.48	4.58	4.40	4.31	4.20	4.28	4.26	4.35
	Zhou et al. (2026) - BIGSTACK_mixed	4.79	4.76	4.27	4.47	4.66	4.90	4.39	4.09	4.88	4.42	4.50	4.57	4.26	4.11	4.24	3.95	4.25	4.35	4.28	4.12	4.03	4.07	3.85	4.18
	Zhou et al. (2026) - BIGSTACK_mixedA	4.88	4.88	4.30	4.52	4.79	5.03	4.58	4.22	5.06	4.56	4.61	4.62	4.28	4.17	4.36	4.08	4.44	4.44	4.45	4.16	4.11	4.16	4.17	4.35
	Zhou et al. (2026) - BIGSTACK_mixedP	4.74	4.68	4.27	4.46	4.55	4.84	4.29	4.03	4.76	4.34	4.37	4.51	4.13	4.04	4.16	3.97	4.19	4.26	4.16	4.02	3.97	3.99	4.00	4.08
ΔSST (°C)	Clark et al. (2024) - Global Mean ΔSST stack	-3.29	-3.07	-1.85	-3.05	-3.12	-3.83	-3.02	-1.41	-2.79	-2.72	-3.87	-3.58	-1.73	-1.87	-1.61	-1.35	-1.83	-1.10	-1.86	-0.98	-0.65	-0.95	-0.37	-0.72
	Clark et al. (2024) - North Atlantic ΔSST stack	-5.81	-5.21	-3.60	-4.73	-6.46	-6.88	-3.91	-1.95	-4.39	-5.71	-7.59	-4.39	-3.48	-5.37	-3.71	-3.76	-4.20	-2.52	-4.63	-2.63	-2.45	-2.09	-2.15	-3.00
	Clark et al. (2024) - Tropical Atlantic ΔSST stack	-3.20	-1.92	-1.36	-2.73	-3.04	-5.20	-4.34	-1.91	-4.25	-3.17	-4.13	-4.32	-2.23	-2.42	-2.90	-0.63	-1.87	-1.79	-2.30	0.70	0.72	-0.48	0.65	-0.60
	Clark et al. (2024) - South Atlantic ΔSST stack	-3.85	-3.75	-2.33	-4.26	-3.53	-5.02	-5.26	-2.31	-4.46	-3.80	-4.06	-3.80	-3.68	-2.69	-3.44	-4.07	-5.64	-4.47	-3.98	-3.71	-0.22	-0.97	-1.79	-1.64
	Clark et al. (2024) - North Pacific ΔSST stack	-2.65	-3.44	-1.29	-2.38	-2.10	-3.42	-3.82	-0.57	-2.49	-1.61	-3.83	-3.64	-0.79	-1.36	-0.20	-1.31	-1.30	-0.67	-0.55	-1.18	0.92	-0.83	0.89	0.41
	Clark et al. (2024) - Tropical Pacific ΔSST stack	-2.39	-1.88	-1.28	-2.03	-2.35	-2.44	-1.01	-0.56	-1.25	-1.83	-1.75	-2.14	-0.75	-1.24	-1.21	-0.82	-1.19	-0.71	-1.11	-0.75	-0.78	-0.93	-0.26	-0.41
	Clark et al. (2024) - South Pacific ΔSST stack	-3.89	-3.85	-2.15	-3.88	-3.93	-4.63	-3.81	-1.43	-5.85	-3.62	-5.18	-4.97	-2.51	-2.48	-3.43	-2.79	-2.84	-2.26	-2.16	-2.18	-2.74	-2.49	-1.29	-1.52
	Clark et al. (2024) - Tropical ΔSST stack	-2.42	-1.74	-1.11	-2.06	-2.19	-2.56	-1.22	-0.38	-1.16	-1.45	-1.97	-2.03	-0.55	-0.80	-0.88	-0.19	-0.84	-0.50	-0.85	-0.15	-0.14	-0.56	0.30	-0.13
	Herbert et al. (2010)	-1.60	-0.64	-0.11	-1.18	-1.54	-1.71	-0.60	0.05	-0.69	-1.07	-1.45	-2.17	-0.41	-0.20	-1.11	-0.63	-1.00	-0.54	-0.94	-0.10	-0.54	-0.73	0.22	0.09
ΔMOT (°C)	Clark et al. (2025)	-3.43	-3.20	-1.98	-3.18	-3.26	-3.97	-3.15	-1.54	-2.92	-2.85	-4.01	-3.72	-1.76	-1.86	-1.53	-1.23	-1.59	-0.91	-1.40	-0.75	-0.50	-0.66	-0.29	-0.46
Mean loess size (normalised)	Ding et al. (2002)	-0.09	-0.24	0.07	0.23	-0.08	0.08	0.62	0.64	0.03	0.35	0.37	-0.27	0.57	0.65	0.68	0.71	0.37	0.45	-0.61	0.42	0.57	0.64	0.63	0.50
Mean size (μm)	Sun et al. (2021)	44.02	46.98	40.73	36.22	44.18	41.24	33.16	33.21	57.72	37.37	35.84	52.94	31.25	33.04	30.94	33.89	53.92	30.86	42.02	33.10	28.89	28.66	32.53	31.86
	Sun et al. (2019)	42.60	43.13	46.70	34.51	47.28	40.84	31.77	31.30	42.62	34.33	37.29	50.64	37.27	34.12	32.61	33.35	42.08	39.89	50.64	37.67	29.36	34.92	38.36	33.17
Inorganic loess carbonates $\delta^{13}\text{C}$ (‰)	Sun et al. (2019)	-0.63	-0.94	-1.21	-1.00	-1.01	-0.89	-1.48	-1.61	-0.89	-0.87	-1.24	-0.93	-1.59	-1.41	-1.62	-2.13	-1.16	-0.82	-1.21	-1.20	-1.69	-1.44	-1.08	-1.67
MAP (mm/day)	Sun et al. (2019)	2.12	2.10	2.18	2.13	2.15	2.16	2.14	2.21	2.10	2.12	2.12	2.11	2.15	2.20	2.13	2.18	2.21	2.16	2.18	2.22	2.22	2.15	2.24	2.14
MAT (°C)	Sun et al. (2019)	9.50	9.00	9.10	9.40	9.50	9.50	9.40	9.60	9.00	9.20	9.80	9.60	9.40	9.60	9.70	10.10	9.80	9.80	9.60	9.70	9.70	9.50	9.90	9.40
	Zhao et al. (2021)	-2.87	-2.58	-3.05	-2.55	-2.54	-3.37	-3.37	-2.42	-2.62	-2.78	-2.64	-2.95	-2.66	-2.18	-1.11	-2.11	-2.50	-2.19	-2.36	-2.45	-2.98	-2.16	-3.11	-3.73
AP (excl. Graminea and aquatic/semi aquatic taxa) (%)	Tzedakis et al. (2006)	2.09	1.35	4.07	0.44	3.21	0.00	13.44	7.89	1.33	2.19	16.30	4.89	6.02	23.41	18.01	3.89	16.37	26.59	7.23	6.09	9.16	nan	nan	nan
AP (excl. pinus) (%)	Donders et al. (2021)	8.00	9.48	15.93	7.89	11.29	1.89	8.70	21.79	10.22	12.44	9.13	4.12	12.69	20.75	15.32	30.12	13.08	35.50	46.25	36.60	14.83	nan	nan	nan
AP (%)	Zhao et al. (2020)	2.39	1.79	2.75	1.99	2.36	1.46	1.44	6.84	2.46	7.29	2.76	5.17	4.98	6.07	18.30	10.05	12.09	2.61	16.36	5.12	9.52	9.09	6.01	4.17
AP (%), excl. Quercus & Alnus	Torres et al. (2013)	17.54	10.29	34.88	11.48	9.33	9.80	6.67	15.57	3.77	2.56	4.95	6.95	17.92	12.50	17.13	21.65	23.23	15.08	10.36	16.29	12.69	14.36	10.99	1.78
Si/Ti	Melles et al. (2012)	0.21	0.27	0.29	0.32	0.36	0.35	0.43	0.38	0.35	0.36	0.41	0.38	0.40	0.45	0.44	0.50	0.41	0.47	0.30	0.43	0.45	0.41	0.41	0.23
Biogenic Silicia (BioSi %)	Prokopenko et al. (2006)	2.00	1.80	2.50	2.70	3.20	2.50	5.20	3.20	3.00	3.30	3.00	3.00	3.40	3.10	2.40	2.80	3.00	3.10	2.80	2.70	3.10	3.10	4.20	3.00
Log Ca/K	Sun et al. (2021)	-1.71	-1.83	-1.56	-1.66	-1.97	-2.12	-1.83	0.51	-2.66	-2.41	-2.41	-2.66	-2.12	-2.12	-2.41	-2.30	-2.53	-2.41	-2.04	-1.90	-1.05	nan	nan	nan